Product, Quotient, and Power Rules for Logarithms

The product rule, the quotient rule, the power rule, and combining all three to condense or expand a logarithm

The three rules (for $b > 0$, $b \neq 1$, and positive inputs)

Rule	Statement	Worked pattern
Product	$\log_b(MN) = \log_b M + \log_b N$	$\log_3 9 + \log_3 3 = \log_3 27 = 3$
Quotient	$\log_b \frac{M}{N} = \log_b M - \log_b N$	$\log_2 40 - \log_2 5 = \log_2 8 = 3$
Power	$\log_b(M^p) = p \log_b M$	$3 \log_5 5 = \log_5 5^3 = 3$

Directions

- Each rule reads in both directions: use it left to right to *expand*, right to left to *condense*.
- Work from the pattern in the table on problems 1–3, then keep going on your own.
- In problems where the base is written as b, the answer holds for every legal base — leave b in your answer.
- Where an exact value exists, give the exact value, not a decimal.

1. Write $\log_2 8 + \log_2 4$ as a single logarithm, then give its exact value.

Answer: ____________

2. Write $\log_3 54 - \log_3 2$ as a single logarithm, then give its exact value.

Answer: ____________

3. Use the power rule to give the exact value of $2\log_4 8$.

Answer: ____________

4. Expand $\log_b(5x)$ as a sum of logarithms.

Answer: ____________

5. Expand $\log_b\left(\frac{x}{7}\right)$ as a difference of logarithms.

Answer: ____________

6. Expand $\log_b(x^5)$ so that no exponent remains inside the logarithm.

Answer: ____________

7. Write $\log_2 3 + \log_2 8 - \log_2 6$ as a single logarithm, then give its exact value.

Answer: ______________

8. Expand $\log_b(x^2y^3)$ completely, so that each logarithm has a single variable inside it and no exponent.

Answer: ______________

9. Suppose $\log_b x = 1.7$ and $\log_b y = 0.4$.

(a) Expand $\log_b\left(\dfrac{x^3}{y}\right)$ completely: ________

(b) Use the given values to find the value of that expression: ________

10. Condense $3\log_b x + \frac{1}{2}\log_b y$ into a single logarithm.

Answer: ________________

11. Powers, exactly and approximately.

(a) Give the exact value of $\log_5(125^2)$: ________

(b) Given $\log_5 3 \approx 0.6826$, find $\log_5 9$ rounded to two decimal places: ________

12. Challenge. Consider $\frac{1}{2}\log_2 36 + \log_2 10 - \log_2 15$.

(a) Write it as a single logarithm, then give its exact value: ________

(b) Explain why the coefficient $\frac{1}{2}$ may be moved inside the logarithm as a square root. Name the rule you are using.